

GROUP 6 – CENTRALIZED LOG MONITORING SYSTEM

Part E: Vulnerabilities, Risks, Recommendations

Executive Summary

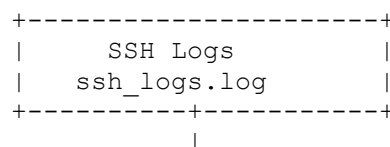
This project developed a centralized log monitoring system using Docker, Python, OpenSearch, and OpenSearch Dashboards. The objective was to collect logs from multiple sources, store them in a centralized repository, and provide visualization for monitoring and security analysis. The system successfully ingests and indexes logs generated from SSH authentication, web server activity, and application events, allowing administrators to detect suspicious activities more efficiently.

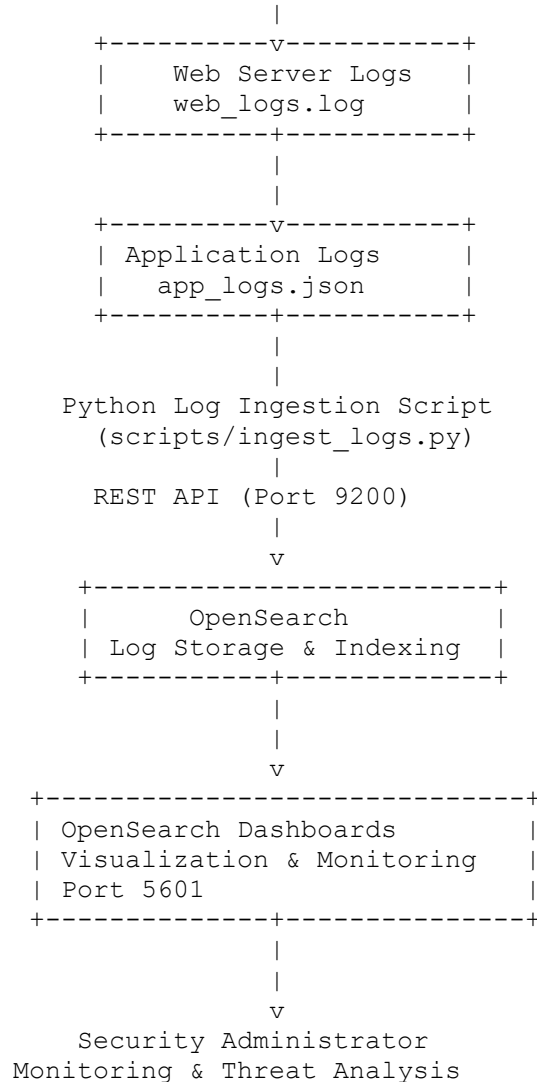
The project demonstrates the importance of centralized logging in improving system monitoring and incident response. By consolidating logs into a single platform, administrators can identify trends, troubleshoot system issues, and investigate security incidents more effectively.

Despite achieving its intended functionality, the security assessment identified several vulnerabilities that could affect the confidentiality, integrity, and availability of the system. These include disabled authentication mechanisms, exposed service ports, lack of encrypted communication, and insufficient access controls. If deployed in a production environment, these weaknesses could expose sensitive log data to unauthorized users and increase the likelihood of cyberattacks.

To improve the overall security posture, several recommendations are proposed, including enabling the OpenSearch security plugin, implementing role-based access control, encrypting network communications using HTTPS, restricting exposed ports, securing Docker containers, and performing regular vulnerability assessments. Implementing these recommendations would significantly improve the security, reliability, and resilience of the centralized log monitoring solution.

Architecture Diagram





Architecture Description

The centralized log monitoring system consists of three primary log sources: SSH authentication logs, web server logs, and application logs. These logs are collected and processed by a Python ingestion script, which sends them to OpenSearch through its REST API. OpenSearch stores and indexes the logs, making them searchable and available for analysis. OpenSearch Dashboards connects to OpenSearch to provide visual dashboards, enabling administrators to monitor system activity, identify suspicious behavior, and investigate security incidents in real time.

Vulnerabilities

The security assessment identified several vulnerabilities within the centralized log monitoring system.

Vulnerability	Description	Severity
Disabled OpenSearch Security Plugin	Authentication and authorization are disabled, allowing unrestricted access to OpenSearch.	High
Disabled Dashboard Authentication	OpenSearch Dashboards can be accessed without user authentication.	High
Exposed Network Ports	Ports 9200, 5601, and 9600 are publicly exposed, increasing the attack surface.	High
Unencrypted Communication	Data is transmitted using HTTP instead of HTTPS, making it vulnerable to interception.	Medium
No Encryption of Stored Logs	Log files stored in OpenSearch are not encrypted.	Medium
No Role-Based Access Control	All users have unrestricted access to system resources.	High
Lack of Log Integrity Verification	No hashing or digital signatures are used to verify that logs have not been modified.	Medium
No Backup Strategy	The system lacks automated backup and disaster recovery mechanisms.	Medium
Resource Exhaustion Risk	Excessive log generation could overload OpenSearch and affect system performance.	Medium
No Monitoring of Log Ingestion Failures	Failures in the ingestion script may go unnoticed due to lack of alerting.	Low

Risks

The identified vulnerabilities introduce several risks that may affect the confidentiality, integrity, and availability of the monitoring system.

Risk	Impact	Likelihood	Risk Level
Unauthorized access to log data	High	High	High
Exposure of sensitive information	High	Medium	High
Modification or deletion of security logs	High	Medium	High
Loss of forensic evidence	High	Medium	High
System downtime due to excessive log traffic	Medium	Medium	Medium
Unauthorized viewing of dashboards	Medium	High	High
Loss of monitoring during ingestion failures	Medium	Medium	Medium
Permanent data loss due to storage failure	High	Medium	High

Recommendations

The following recommendations are proposed to improve the security and reliability of the centralized log monitoring system.

1. Enable OpenSearch Security

Enable the OpenSearch Security Plugin to provide user authentication and authorization. This prevents unauthorized access to log data and dashboards.

2. Enable HTTPS

Configure SSL/TLS certificates to encrypt communication between the ingestion script, OpenSearch, and OpenSearch Dashboards.

3. Implement Role-Based Access Control

Assign different permissions to administrators, analysts, auditors, and read-only users to ensure users only access resources required for their responsibilities.

4. Restrict Network Access

Limit access to ports 9200, 5601, and 9600 using firewall rules or Docker network policies. Only trusted devices should communicate with the monitoring system.

5. Encrypt Stored Logs

Enable encryption for storage volumes or configure encrypted OpenSearch indices to protect sensitive information if storage devices are compromised.

6. Protect Log Integrity

Use cryptographic hashing or digital signatures to verify that log files have not been altered or deleted.

7. Implement Automated Backups

Schedule regular backups of OpenSearch indices to prevent data loss and support disaster recovery.

8. Configure Monitoring and Alerts

Implement automated alerts for failed log ingestion, unusual login attempts, high CPU usage, low disk space, and service failures.

9. Secure Docker Containers

Follow Docker security best practices by:

- Running containers as non-root users.
- Keeping container images updated.
- Removing unnecessary services.
- Applying the principle of least privilege.

10. Conduct Regular Security Assessments

Perform periodic vulnerability scanning and penetration testing using tools such as OpenVAS, Nessus, or Trivy to identify and remediate newly discovered security weaknesses.

Conclusion

The centralized log monitoring system successfully demonstrates how logs from multiple sources can be collected, stored, and visualized using OpenSearch and OpenSearch Dashboards. The system provides a practical platform for monitoring security events and improving incident response. However, the assessment identified several vulnerabilities, including disabled security features, exposed network ports, and inadequate access controls, which increase the overall security risk. By implementing the recommended security measures, such as enabling authentication, encrypting communications, restricting access, and performing regular security assessments, the system can be significantly strengthened and made suitable for deployment in a production environment.